

DRAFT COVER LETTER

NOT READY FOR SUBMISSION - PRE-OPEN / INCOMPLETE

Status: Internal preparation draft; do not send to a journal

Manuscript: ThermoRoute retrospective daily river-temperature hindcasting

Date: 22 July 2026

Byte-frozen pre-opening snapshot: this file is not a live status page and is intentionally not rewritten by the one-time Route-A result renderer.

[Date]

Dear Editor,

We are preparing a manuscript entitled “ThermoRoute: a bounded-deviation, physics-inspired framework for retrospective daily river-temperature hindcasting.” The submission package is not yet complete, and this draft should not be sent to a journal.

The planned study evaluates daily water-temperature hindcasts at 1-, 3-, and 7-day horizons on a stable 120-site USGS development panel. The method combines a damped-persistence anchor, a learned relaxation proposal, a sparse variable/lag router, a causal temporal convolutional encoder, a regime mixture, bounded residuals, a separate MSE point head, three pinball-trained quantile heads, and a split-conformal 90% interval. The evaluation design emphasizes common target keys, station-balanced effects, clustered spatial inference, frozen model/input chronology, raw qualifier provenance, and receipt-derived statements. The H2 comparison is limited to the frozen four-candidate, five-seed LightGBM procedure; historical tuning budgets were not equalized. The Stage 09 controls are single-seed functionality/intervention diagnostics rather than evidence of causal mechanisms, component necessity, or cross-seed stability.

An outcome-free inference-scope amendment makes all directional wording conditional on a claim-blocking gate requiring at least 30 reportable clusters. The frozen cohort contains at most 15 HUC2 groups, so this requirement cannot pass regardless of the target-period outcomes. The five comparisons will still be reported, but only as fixed-cohort descriptive effects; their HUC2 bootstrap intervals, exact sign-flip p-values, and Holm values are assumption-conditional sensitivities and cannot support superiority, non-inferiority, equivalence, parity, or a U.S.-river superpopulation claim.

The outcome-free 2018–2020 predictor-product bridge has passed. At present, the canonical development rerun, model-suite freeze, later covariate freeze, one-time authorization, and target-period scoring have not finished. We have therefore removed all legacy performance numbers from the manuscript. A future submission will include every predeclared formal comparison, including negative, conflicting, or non-estimable outcomes, only after the fixed evidence chain passes.

The intended contribution is twofold: a constrained hindcasting architecture and an auditable evaluation/release workflow. The manuscript will explicitly delimit the availability-enriched sample, retrospective finalized covariates, coarse HUC2 dependence groups, daily-mean target, point-scale meteorology, missing development request provenance, numerical comparison margin, calibration-row-weighted Platt fitting versus station-balanced target metrics, observable-outcome conditioning, the descriptive-only temporal-coverage sensitivities, the unofficial status of the Air2stream-style reference, and the local honest-owner Git seal.

Before submission we must also replace all author, affiliation, ORCID, e-mail, funding, competing-interest, repository, DOI, and data-license placeholders with verified information.

Sincerely,

[Corresponding author — name and verified contact details required]